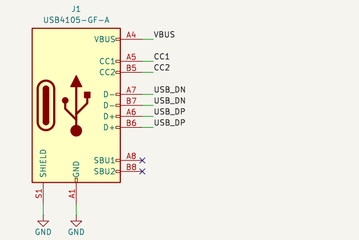
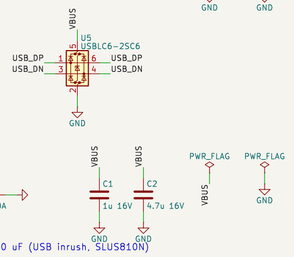
1 USB-C input + ESD

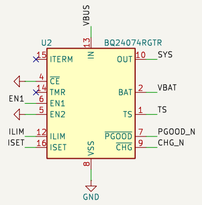




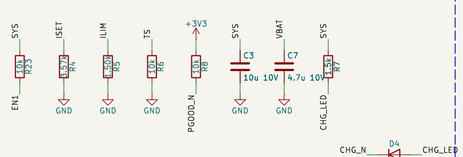


IN total <= 10 uF (USB inrush, SLUS810N)

2 BQ24074 charger + power path

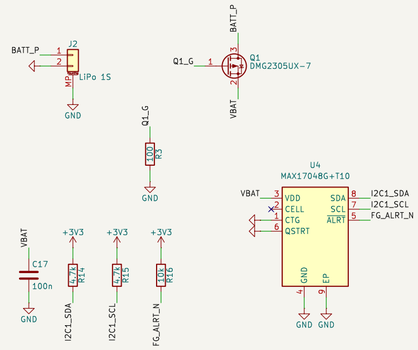


TMR open: default timers (30 min / 5 h). ITERM open: 10 % termination.
EN2=0 EN1=1: 500 mA USB input limit. -CE=0: charging on.



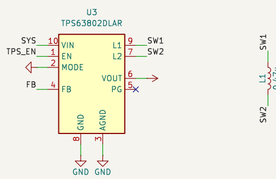
R4: Ichg = 890/3570 = 0.25 A. R5: Ilim = 1610/1500 = 1.07 A. R6: TS fixed (no NTC).

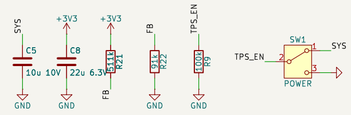
3 Battery, reverse-polarity FET, fuel gauge



Body diode conducts first, then Vgs = −Vbat turns Q1 on. Reverse cell: off.

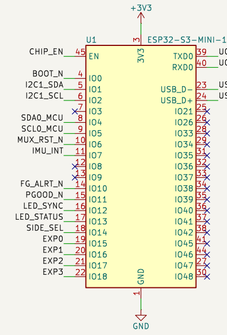
4 TPS63802 3.3 V buck-boost

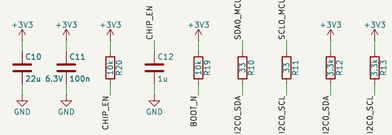


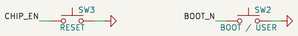


VOUT = 0.5 V x (1 + 511k/91k) = 3.31 V. MODE=0: power save. PG unused.

5 ESP32-S3-MINI-1-N8



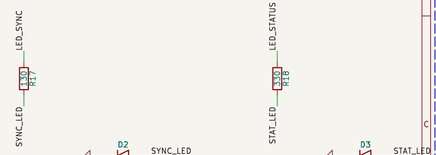


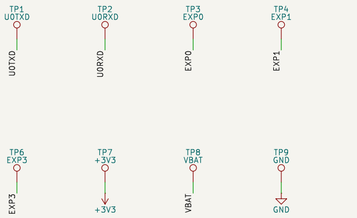


GPIO3/45/46 straps: NC. GPIO19/20 = native USB. GPIO26+ (flash) and spare IOs: NC.

6 Plate link, LEDs, test points

Pin 1 = GND on every harness. I2C0 400 kHz, 3.3k here + 3.3k per channel on the plate.





Open items: see design.md section 5
Pinouts and values checked against TI SLVSEU9D, SLUS810N, Maxim 19-6171
Generated by kicad/tools/gw_pod.py — edit the script, not the sheet
ESP32-S3-MINI-1-N8, BQ24074 power path, TPS63802 3.3 V, MAX17048
Tendra Hand — open source (CERN-OHL-S-2.0)

Sheet: /
File: glove_pod.kicad_sch

Title: Tendra Glove — GW-POD wrist pod

Size: A2	Date: 2026-11-14	**Rev: A0 (draft, not built)**
KiCad E.D.A. 9.0.9		Id: 1/1